## Lab 2 Specifications

The lab 2 specifications are trying to get you to focus on two new topics: constructing large Verilog systems by intelligently reusing modules, and clean schematic construction.

### Hardware Specifications

#### Proficiency

- ☐ Two seven-segment displays show two independent hexadecimal numbers.
- ☐ Resistors are in series with FPGA pins so that they correctly limit current.
- ☐ Scanning signal appears on LEDs when appropriate buttons are depressed on the keypad.

#### Excellence

- ☐ Seven segment displays are same brightness regardless of how many segments are lit
- ☐ No noticeable bleeding of the digits between displays.
- ☐ No flickering on the individual digits.
- ☐ Scanning signal continues to be displayed under different keypress conditions, including multiple touches on rows or columns.

### Verilog Structure and Style Specifications

#### Proficiency

- ☐ HDL design includes only a single seven-segment decoder module that is time multiplexed.
- ☐ Top design only includes instances of HSOSC, seven-segment module, counter module, scanning module and one-line assign statements. The assign statements should implement multiplexing and LED pass through.
- ☐ Scanning module only includes instances of a counter and assign statements that convert from the counter output to output values.
- ☐ Scanning module is resettable and enable-able.
- ☐ No counters implemented inline, all of them are contained in modules.
- ☐ Style conventions from previous labs (module names, descriptive variables, etc.) are still followed. See HDL specs.

#### Excellence

- ☐ All counter modules (scanning and multiplexing) are parametrized instances of the counter module from lab 1. (Minor adjustments, especially to respond to instructor feedback, are allowed. But any changes to the Verilog need to be verified by re-running the testbench.)

### Simulation Specifications

#### Proficiency

- ☐ No repeat simulations from lab 1. Those modules have already been verified! (But do include links to the lab 1 simulations in your report. Also, as above, include sims verifying modules that you changed.)
- ☐ QuestaSim simulations carried out (possibly manually) for the scanning module and the top module.
- ☐ At least one simulation uses an automatic testbench. (Lab 3 will require automatic testbenches for all modules.)
- ☐ All simulations captured as legible waveform screenshots in report.

#### Excellence

- ☐ Automatic testbench for the scanning module that demonstrates all four output transitions, enable and reset features.
- ☐ Automatic testbench for the top module that exercises multiplexing functionality and LED driving functionality.

### Block Diagram Specs

#### Proficiency and Excellence

- ☐ Block diagram exists and a good faith effort was made to follow the specs below. Block diagram specs in full effect in lab 3.

### Other Documentation

#### Proficiency

- ☐ Current draw/sink on all FPGA pins are below the currents specified in the recommended operating conditions. Claims are backed up by calculations and reference to the appropriate items on the datasheet.

#### Excellence

- ☐ No errors in the calculations (correct forward voltage read off the datasheet, proper voltages used, etc.)

### Schematic Specifications

#### Proficiency

- ☐ Schematic correctly represents the circuit that was built on the breadboard.
- ☐ No parts of motherboard schematic replicated. Shows only off-board components. Indicate on-board components with a brief paragraph that links the motherboard schematic (here).
- ☐ No parts internal to the chip (e.g., internal pullup resistors) drawn on schematic. Indicate important pullups, capacitors, etc. with comments instead.
- ☐ All pin names labeled.
- ☐ All pin numbers labeled.
- ☐ Crossing wires clearly identified as junction or unconnected.
- ☐ Neat layout (e.g., clear organization and spacing).
- ☐ All parts labeled with part number.
- ☐ All component values present.

#### Excellence

- ☐ Standard symbols used for all components where applicable.
- ☐ Signals "flow" from left to right where possible (e.g., inputs on left hand side, outputs on right hand side, power on top, ground on bottom)
- ☐ Title block with author name, title, and date.

### Lab Writeup

#### Proficiency and Excellence

- ☐ Documentation embedded in a well-structured, concise report with one section per major deliverable. Sections should match structure of other specs.
- ☐ Spoiler tags are used to hide screenshots and code. Some unspoilered entries (~1 waveform, essential code snippets) allowed if they are essential to the narrative, kept to a minimum.
- ☐ Writeup contains minimal spelling or grammar issues and any errors do not significantly detract from clarity of the writeup.
- ☐ Statement of whether the design meets all the requirements. If not, list the shortcomings.
- ☐ AI prototype attempted and some reflection is recorded.
- ☐ Number of hours spent working on the lab are included.
- ☐ (Optional) List comments or suggestions on what was particularly good about the assignment or what you think needs to change in future versions.

## FOR REFERENCE, NOT YET IN EFFECT

### Block Diagram Specifications

### Proficiency

- ☐ Each module is represented by a block diagram
- ☐ Bare logic in all module is represented by appropriately drawn gates.
- ☐ Submodules in each module are represented by blocks inside of the instantiating module.
- ☐ Submodules have all ports represented by pin symbols on the boundary of the submodule, each with a port name.
- ☐ Submodules indicate the name of the module being instantiated and the name of the instance.
- ☐ All connections between module ports, logic and submodules are indicated.
- ☐ Module ports are represented by pin symbols on the boundary of the module and labeled with port names
- ☐ All nets internal to the module are labeled, even when they share a name with a port on a module/submodule boundary.

### Excellence

- ☐ Appropriate use of bus notation for multi-bit signals (e.g., `data[7:0]` for an 8-bit bus), including fan-in and fan-out when needed.
- ☐ Appropriate use of implicit connections (i.e.: connect-by-label-name) to reduce routing connection
- ☐ Block diagram is an exact, 1:1 representation of the Verilog code.